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Level 2 — 5 marks/question

L2-Q1. (M) Repeated

P

CODE

```
def findDuplicate(nums):
    slow = nums[0]
    fast = nums[0]
    while True:
        slow = nums[slow]
        fast = nums[nums[fast]]
        if slow == fast:
            break
    slow = nums[0]
    while slow != fast:
        slow = nums[slow]
        fast = nums[fast]
    return slow

nums = [1,3,4,2,2]
print(findDuplicate(nums))
```

INPUT

nums = [1,3,4,2,2]

OUTPUT

2

L2-Q2. (M) Sort

P

CODE

```
arr = [90,-90,80,66,78,10]

arr.sort()
print(arr)

n = len(arr)
middle = arr[n // 2 - 1] if n % 2 == 0 else arr[n // 2]
print("Middle element=", middle)
```

OUTPUT

```
[-90, 10, 66, 78, 80, 90]
Middle element= 66
```

L2-Q3. (M) Replace

P

CODE

```
S1 = "Hai Welcome to learn coding. It is fun to learn!!!"
S2 = "learn"
S3 = "teach"

result = S1.replace(S2, S3)
print(result)
```

OUTPUT

Hai Welcome to teach coding. It is fun to teach!!!

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L2-Q4. (M) Perfect Square

P

CODE

```
def numsquares(n):  
    squares = [i*i for i in range(1, int(n**0.5)+1)]  
    queue = [n]  
    steps = 0  
  
    while queue:  
        next_queue=[]  
        for num in queue:  
            if num == 0:  
                return steps  
            for sq in squares:  
                next_queue.append(num - sq)  
        queue = list(set(next_queue))  
        steps += 1  
  
print(numsquares(12))
```

OUTPUT

3